Prepared by Er. Shiva Shrestha

Advanced Python Programming For Data Science

Chapter 5: Data Visualization and Storytelling

Chapter Overview

This chapter focuses on:

- Understanding the role of data visualization in data science
- Designing clear and effective visual representations of data
- Visualizing trends, distributions, and relationships
- Using visualization tools to support data-driven decisions

The emphasis is on **communicating insights effectively**, not just creating charts.

Duration: 7 hours

Marks Weightage: 9 Marks

Learning Outcomes

After completing this chapter, students will be able to:

- Explain principles of effective data visualization and dashboard design
- Create basic visualizations using Matplotlib
- Apply Seaborn for statistical and exploratory visual analysis
- Develop interactive visualizations using Plotly
- Interpret visual patterns to generate meaningful insights
- Present data-driven stories for technical and non-technical audiences

Chapter Structure

- **5.1** Principles of effective visualization and dashboard design
- **5.2** Visualization with Matplotlib: Line, bar, histogram, scatter, subplots
- **5.3** Seaborn for statistical visualization: Box plot, pair plot, heat map
- **5.4** Interactive visualization using Plotly
- **5.5** Visualization driven insight generation
- **5.6** Case study: End-to-end visualization and reporting project

5.1 Principles of Effective Visualization and Dashboard Design

Principles of Effective Visualization

Effective visualization aims to:

- Communicate data clearly and accurately
- Highlight important patterns and trends
- Support comparison and decision-making

Key Design Principles

- **Clarity:** Avoid clutter and unnecessary elements
- **Accuracy:** Use correct scales and proportions
- **Simplicity:** Prefer simple charts over complex ones
- **Consistency:** Maintain uniform colors, labels, and styles

Poor visualization can **mislead users**, even if the data is correct.

Steps for Effective Data Visualization

Step 1: Define the Objective
- Identify the question to be answered
- Clarify what decision the visualization should support

Step 2: Understand the Data
- Identify data types (categorical, numerical, time-series)
- Check data quality, scale, and range

Step 3: Choose the Appropriate Chart
- Trend → Line chart
- Comparison → Bar chart
- Distribution → Histogram / Box plot
- Relationship → Scatter plot
- Correlation → Heat map

Step 4: Design the Visualization
- Add clear titles, labels, and legends
- Use consistent scales and limited colors
- Avoid clutter and unnecessary elements

Step 5: Highlight Key Insights
- Emphasize important patterns or anomalies
- Use annotations or color emphasis if needed

Step 6: Interpret and Communicate
- Explain what the visualization shows
- Connect insights to conclusions or actions

Dashboard Design Basics

A good dashboard:

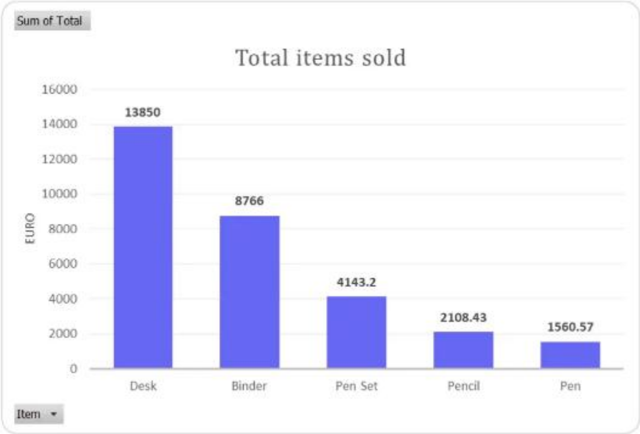
- Answers specific business or analytical questions
- Displays key metrics at a glance
- Uses minimal but meaningful visuals

Common Mistakes

- Too many charts on one screen
- Mixing unrelated metrics
- Overuse of colors and effects

Cluttered chart

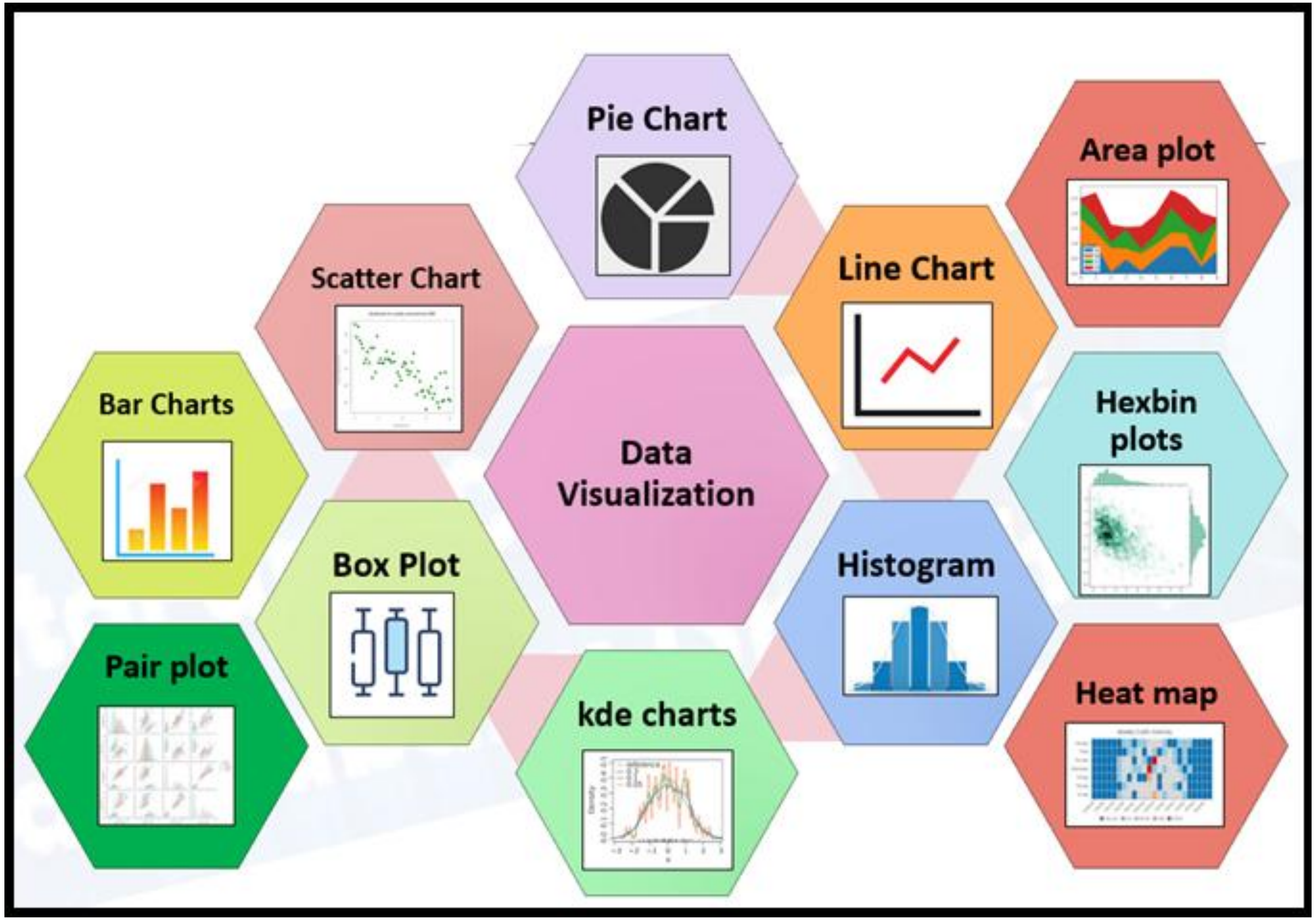
(bad data visualization example)



Clean chart

(good data visualization example)

5.2 Visualization with Matplotlib: Line, bar, histogram, scatter, subplots



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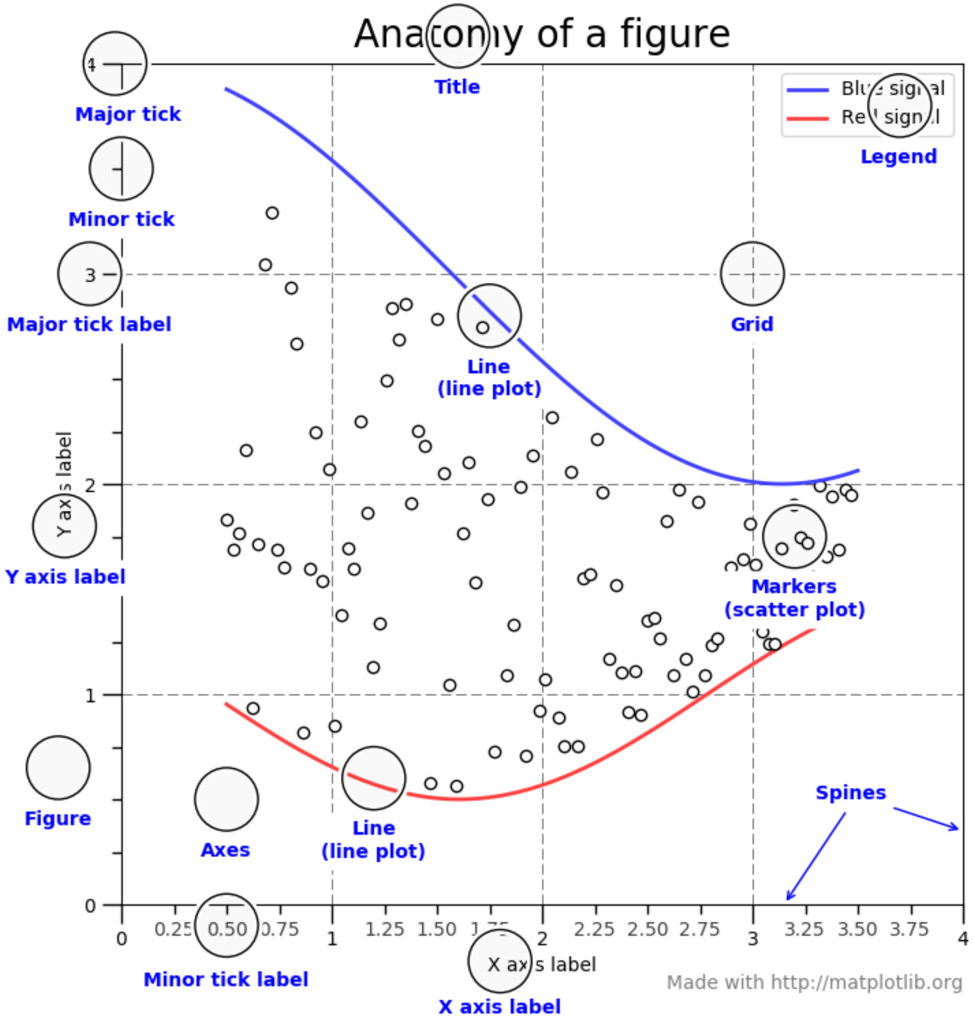
Introduction to Matplotlib

Matplotlib is a core Python library used for:

- Creating **static visualizations**
- Full control over plot elements
- Producing publication-quality charts

Why Matplotlib?

- Flexible and customizable
- Foundation for Seaborn and other libraries
- Widely used in data science and research



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Line Plot

Purpose

- Displays **trends over time**
- Shows changes in continuous data

When to Use

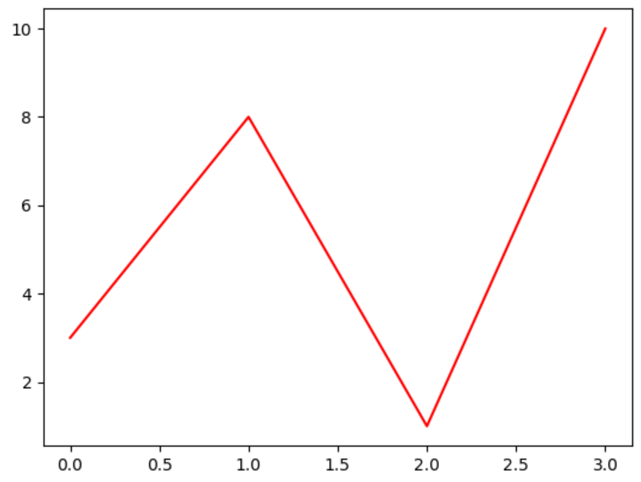
- Time-series data (sales, temperature, stock prices)
- Comparing trends across variables

Key Characteristics

- X-axis: time or ordered values
- Y-axis: continuous numerical values
- Emphasizes **direction and rate of change**

Example Insight

- Upward trend → growth
- Downward trend → decline



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Bar Chart

Purpose

- Compares values across **categories**

When to Use

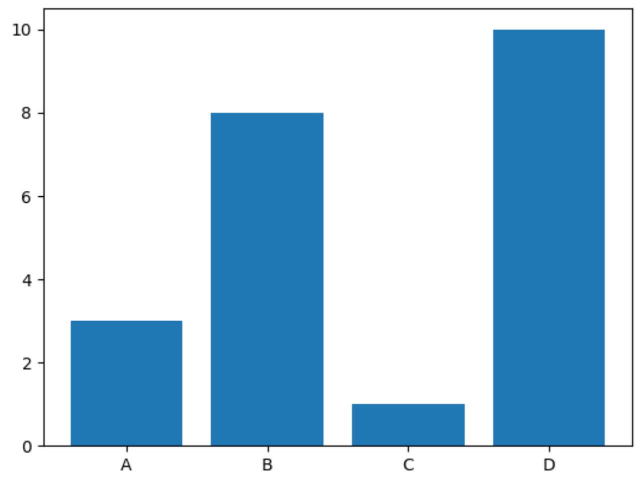
- Department-wise sales
- Product comparison
- Survey results

Key Characteristics

- X-axis: categories
- Y-axis: numerical values
- Height of bars represents magnitude

Best Practice

- Use consistent bar widths
- Avoid too many categories in one chart



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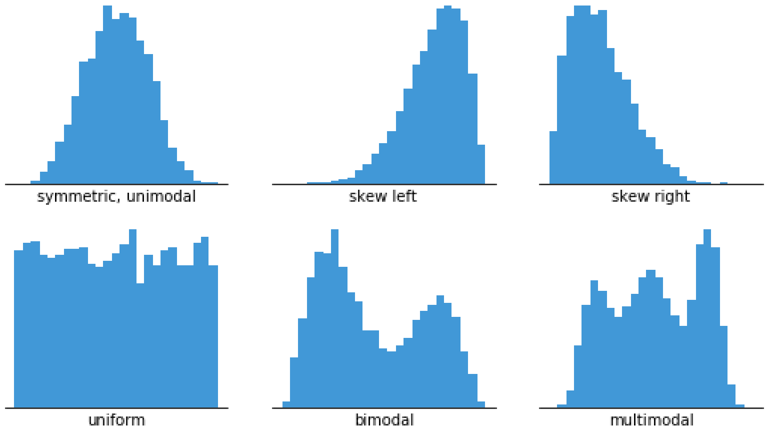
Histogram

Purpose

- Shows **distribution of data**

When to Use

- Understanding data spread
- Identifying skewness and outliers
- Analyzing frequency patterns

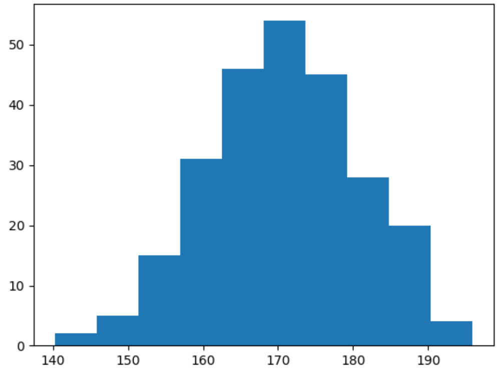


Key Characteristics

- Data divided into **bins**
- Y-axis shows frequency
- Helps detect normal vs skewed distributions

Example Insight

- Right-skewed → majority of lower values
- Wide spread → high variability



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Scatter Plot

Purpose

- Displays **relationship between two variables**

When to Use

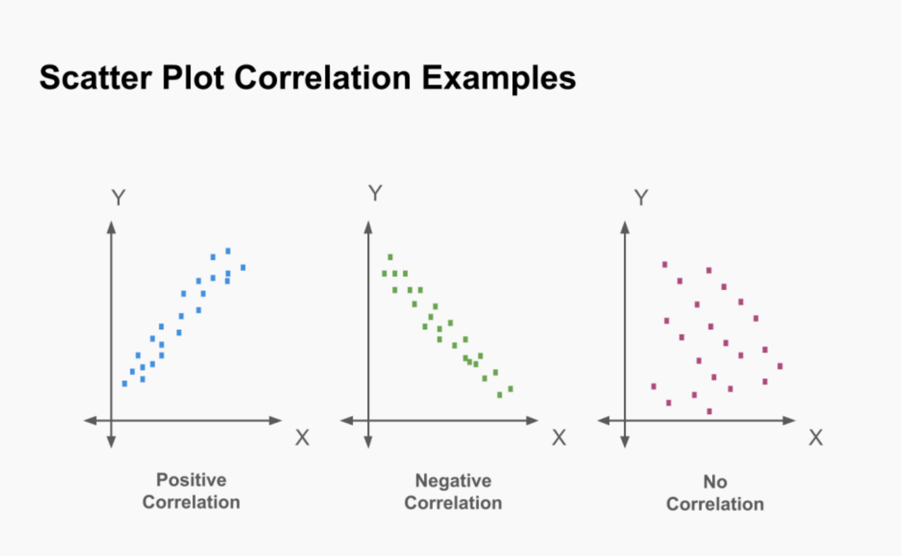
- Correlation analysis
- Feature relationship exploration

Key Characteristics

- Each point represents one observation
- Pattern indicates relationship:
  - Positive correlation
  - Negative correlation
  - No correlation

Example Insight

- Clustering → similar behavior
- Outliers → anomalies



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Subplots

Purpose

- Displays **multiple plots in a single figure**

When to Use

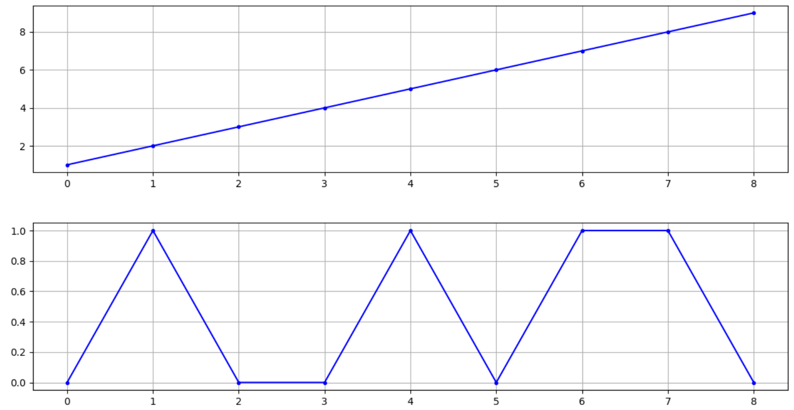
- Comparing different visualizations
- Viewing multiple aspects of data simultaneously

Key Characteristics

- Grid-based layout (rows × columns)
- Shared or independent axes
- Improves comparative analysis

Example Use Case

- Sales trend, distribution, and comparison shown together



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Choosing the Right Plot

Data Requirement	Recommended Plot
Trend over time	Line plot
Category comparison	Bar chart
Data distribution	Histogram
Variable relationship	Scatter plot
Multiple views	Subplots

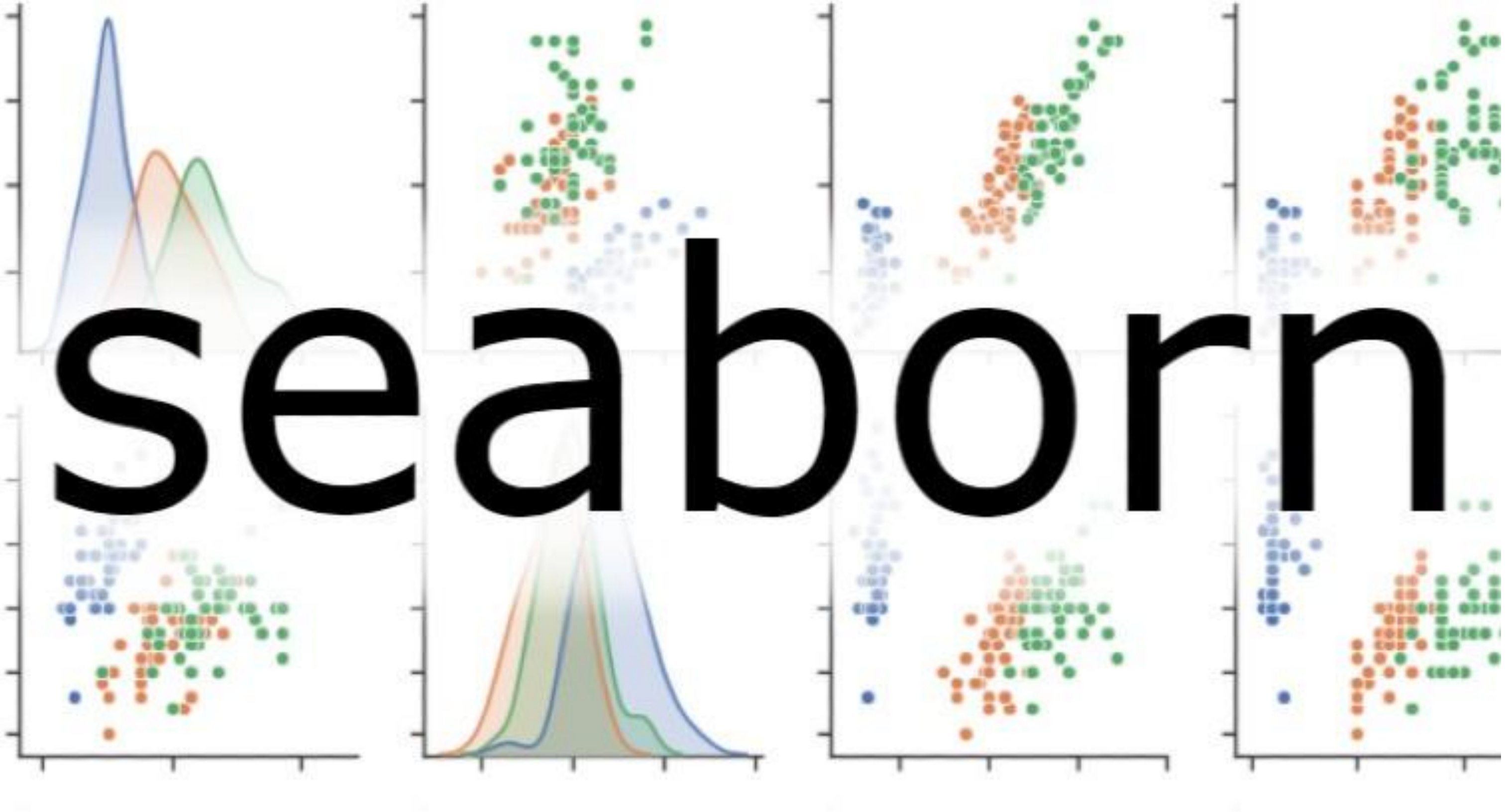
Common Mistakes in Matplotlib Visualizations

- Using wrong chart type
- Missing labels and titles
- Overcrowding plots
- Misleading axis scales

Golden Rule

- The chart type must match the **data and the question**, not convenience.

5.3 Seaborn for statistical visualization: Box plot, pair plot, heat map



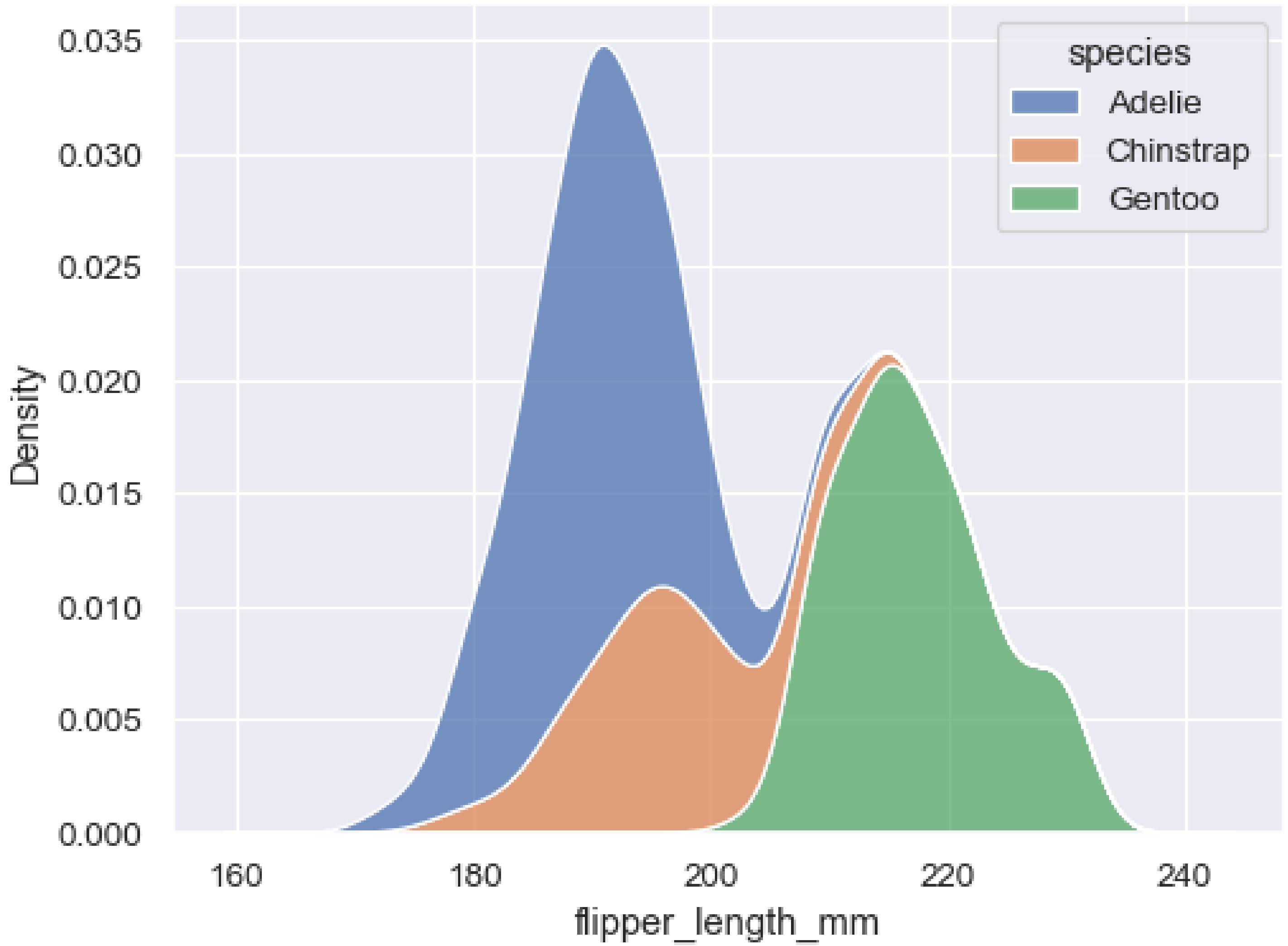
Introduction to Seaborn

Seaborn is a Python library used for:

- Statistical and exploratory data visualization
- High-level interface built on Matplotlib
- Automatic handling of data frames

Why Seaborn?

- Cleaner default aesthetics
- Built-in support for statistical plots
- Ideal for Exploratory Data Analysis (EDA)



Aspect	Matplotlib	Seaborn
Level	Low-level plotting library	High-level statistical plotting (built on Matplotlib)
Defaults	Basic, requires styling	Polished, attractive defaults
Best for	Full control, custom plots	Quick statistical visuals and comparisons
Data input	Works with arrays/lists; Pandas ok	Designed to work seamlessly with Pandas DataFrames
Plot types	Very broad; you build anything	Focused on statistical plots (e.g., box, violin, pair, heatmap)
Code length	More verbose	More concise
Customization	Maximum control	Good, but often through Matplotlib underneath
Learning curve	Steeper	Easier for quick results

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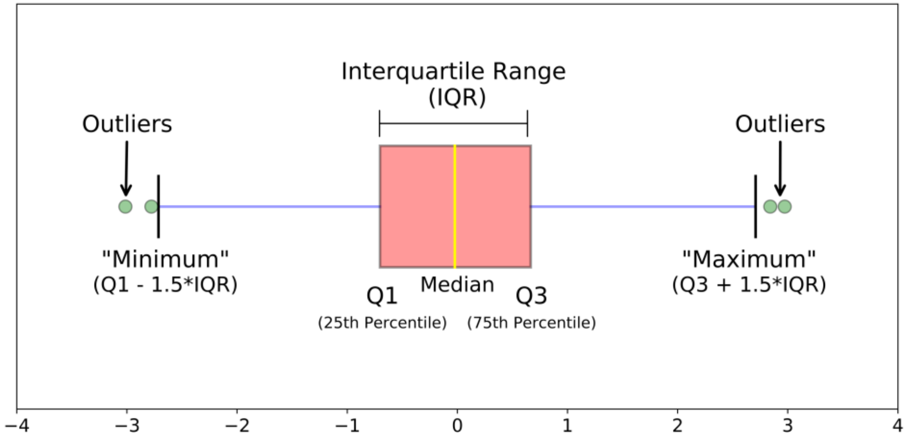
Box Plot

Purpose

- Displays **data distribution and spread**
- Identifies **outliers**

Key Components

- Median (central line)
- Quartiles (Q1 and Q3)
- Interquartile Range (IQR)
- Whiskers and outliers



When to Use

- Comparing distributions across categories
- Detecting variability and anomalies

Interpretation

- Wider box → higher variability
- Outliers → unusual observations

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Pair Plot

Purpose

- Visualizes **pairwise relationships** among multiple variables
- Combines scatter plots and distributions

Key Characteristics

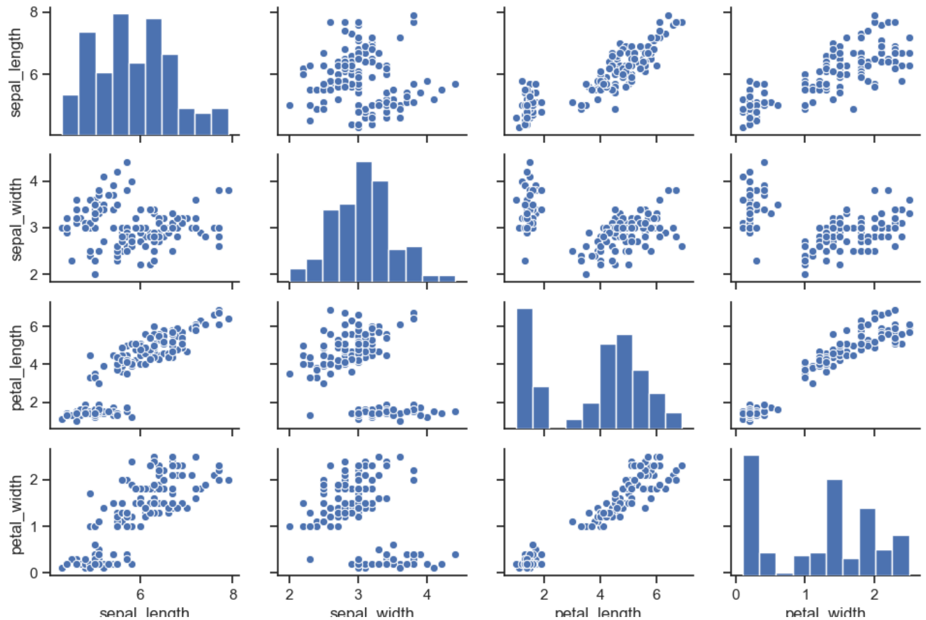
- Diagonal → variable distributions
- Off-diagonal → pairwise scatter plots

When to Use

- Multivariate exploratory analysis
- Identifying correlated variables

Interpretation

- Linear pattern → correlation
- Clusters → groups in data



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Heat Map

Purpose

- Represents data values using **color intensity**
- Commonly used for **correlation matrices**

Key Characteristics

- Color scale shows magnitude
- Darker colors → stronger values

When to Use

- Feature correlation analysis
- Pattern recognition in large matrices

Interpretation

- Strong positive correlation → high positive value
- Strong negative correlation → high negative value

	User Rating	Reviews	Price	Year
User Rating	1.000000	-0.001729	-0.133086	0.242383
Reviews	-0.001729	1.000000	-0.109182	0.263560
Price	-0.133086	-0.109182	1.000000	-0.153979
Year	0.242383	0.263560	-0.153979	1.000000

Choosing the Right Seaborn Plot

Requirement	Recommended Plot
Distribution comparison	Box plot
Multivariate relationships	Pair plot
Correlation strength	Heat map

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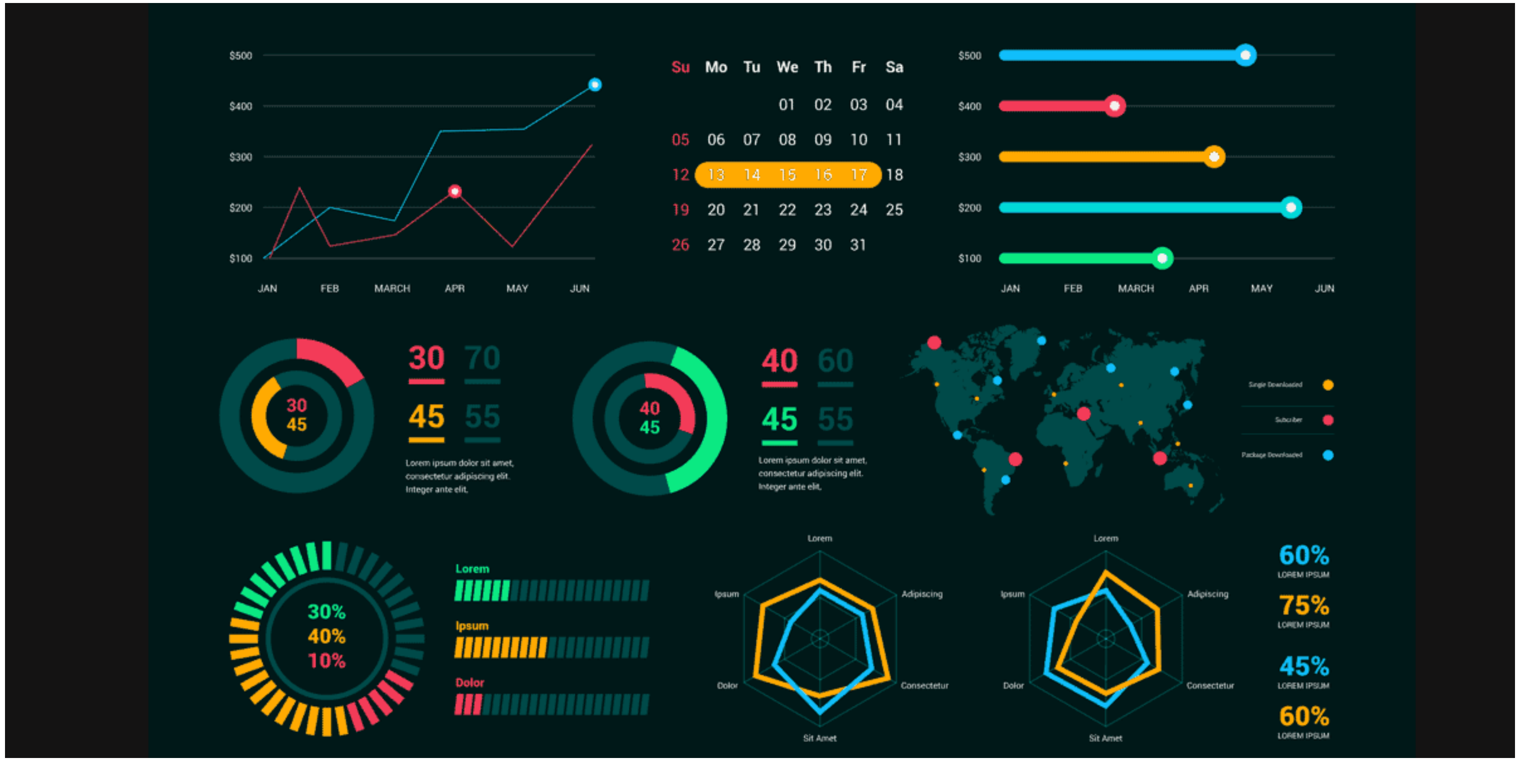
Common Mistakes in Seaborn Visualizations

- Using pair plots on very large datasets
- Ignoring color scale interpretation in heat maps
- Misinterpreting outliers

Best Practice

- Always interpret statistical plots along with summary statistics.

5.4 Interactive Visualization Using Plotly



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Introduction to Interactive Visualization

Interactive visualization allows users to:

- Explore data dynamically
- Interact with charts instead of passively viewing them
- Gain deeper insights through user-driven exploration

Key Difference

- Static visualization → fixed view
- Interactive visualization → user-controlled view

Why Interactive Visualization is Needed

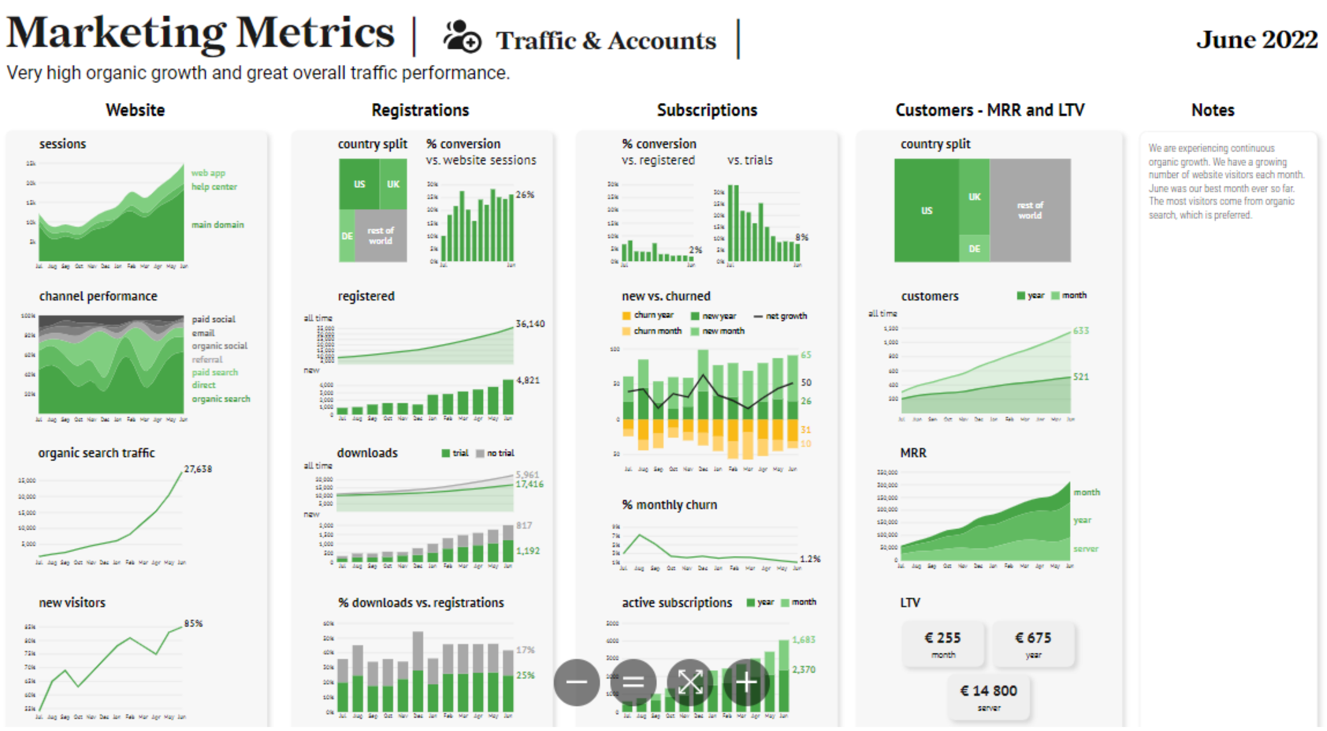
Limitations of Static Charts

- Cannot zoom into details
- No additional information on demand
- Limited user engagement

Advantages of Interactive Charts

- Hover to view exact values
- Zoom and pan to explore patterns
- Toggle data series on/off

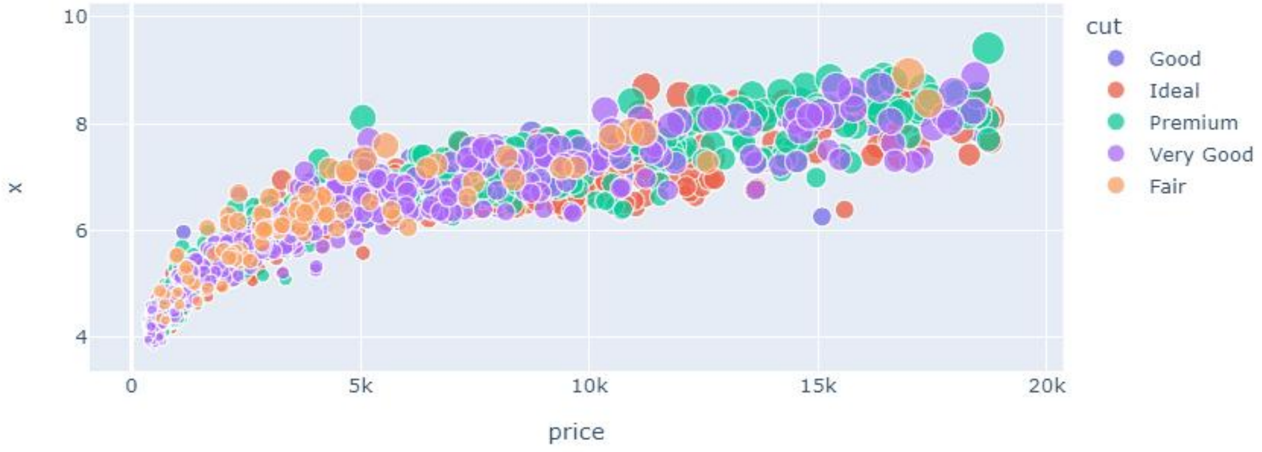
Interactive visualizations are preferred in **business dashboards and analytics tools**.



Introduction to Plotly

Plotly is a Python library used to:

- Create interactive, web-ready visualizations
- Build dashboards and reports
- Support exploratory data analysis



Key Features

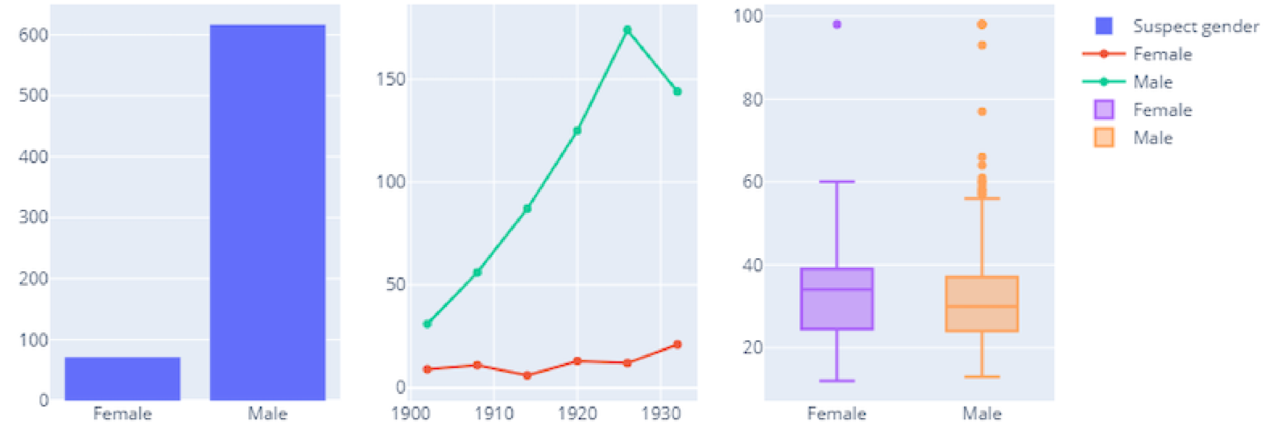
- High-quality interactive charts
- Built-in interactivity without complex code
- Integration with web applications

Common Interactive Features in Plotly

Interactive Capabilities

- **Hover:** Displays detailed data values
- **Zoom:** Focus on specific regions
- **Pan:** Navigate across the chart
- **Legend Interaction:** Show or hide data series

These features help users **discover insights independently**.



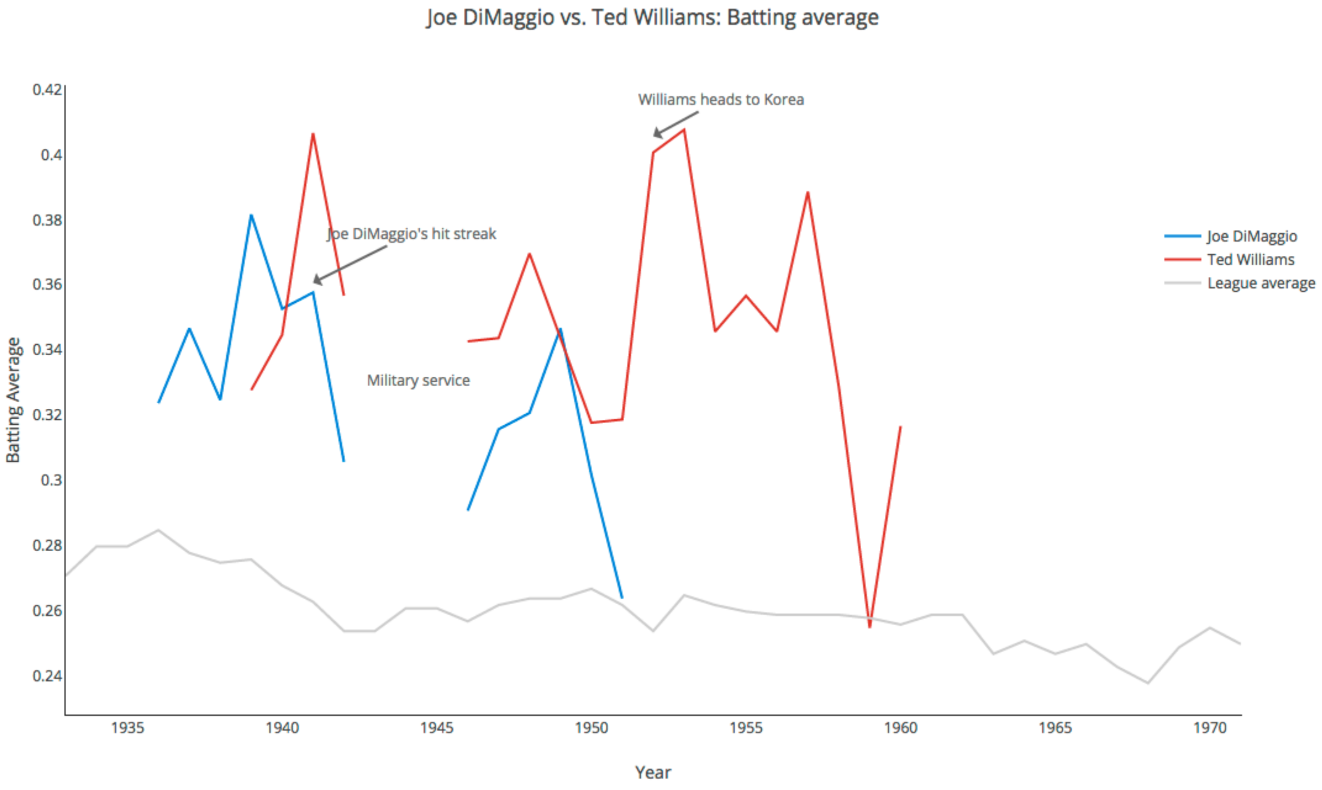
Interactive Line and Bar Charts

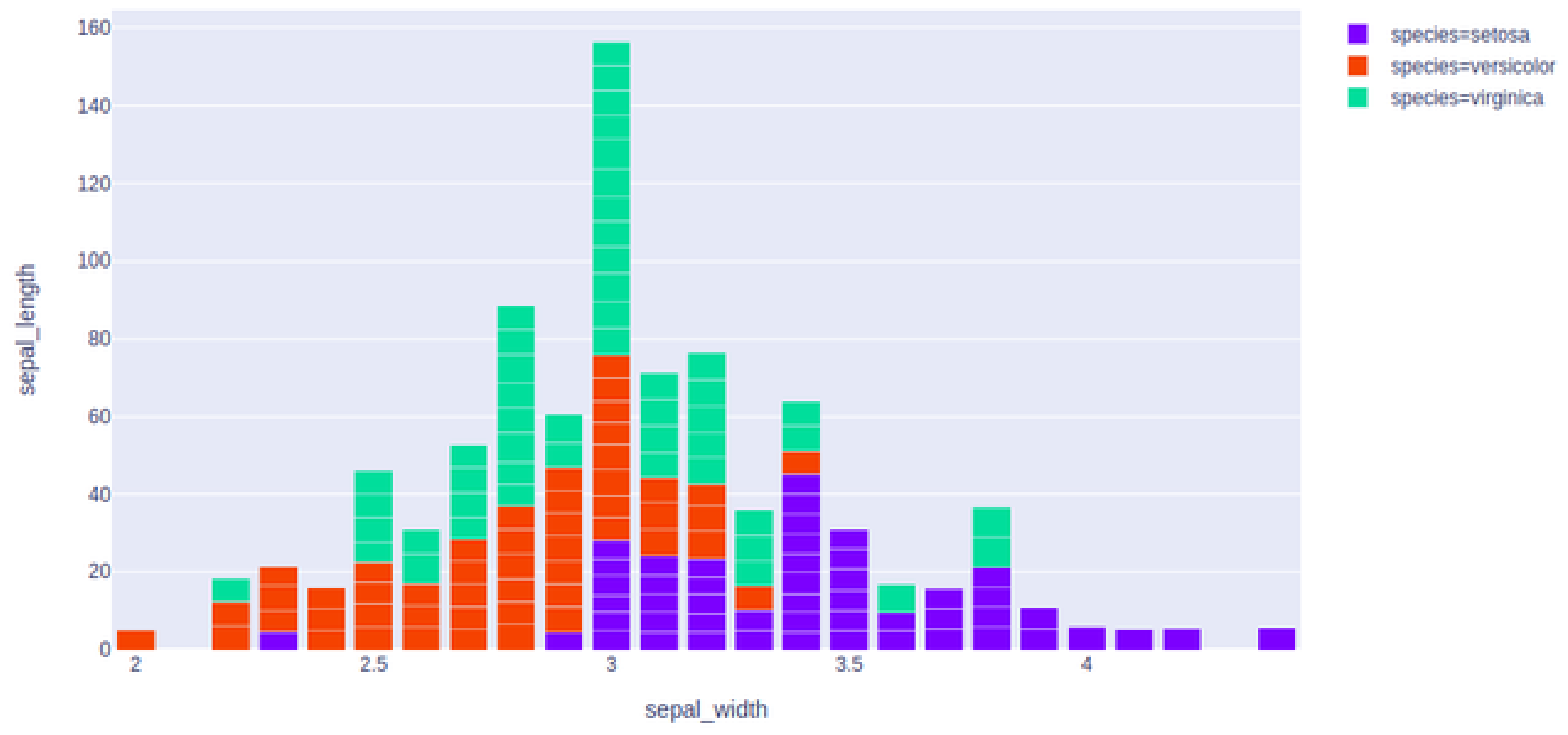
Use Cases

- Time-series trend analysis
- Sales and performance dashboards

Benefits

- Compare trends dynamically
- Identify exact peak and drop points
- Improve clarity for decision-makers





160
140
120
100
80
60
40
20
0
sepal_length
2
2.5
3
3.5
4
sepal_width
species=setosa
species=versicolor
species=virginica

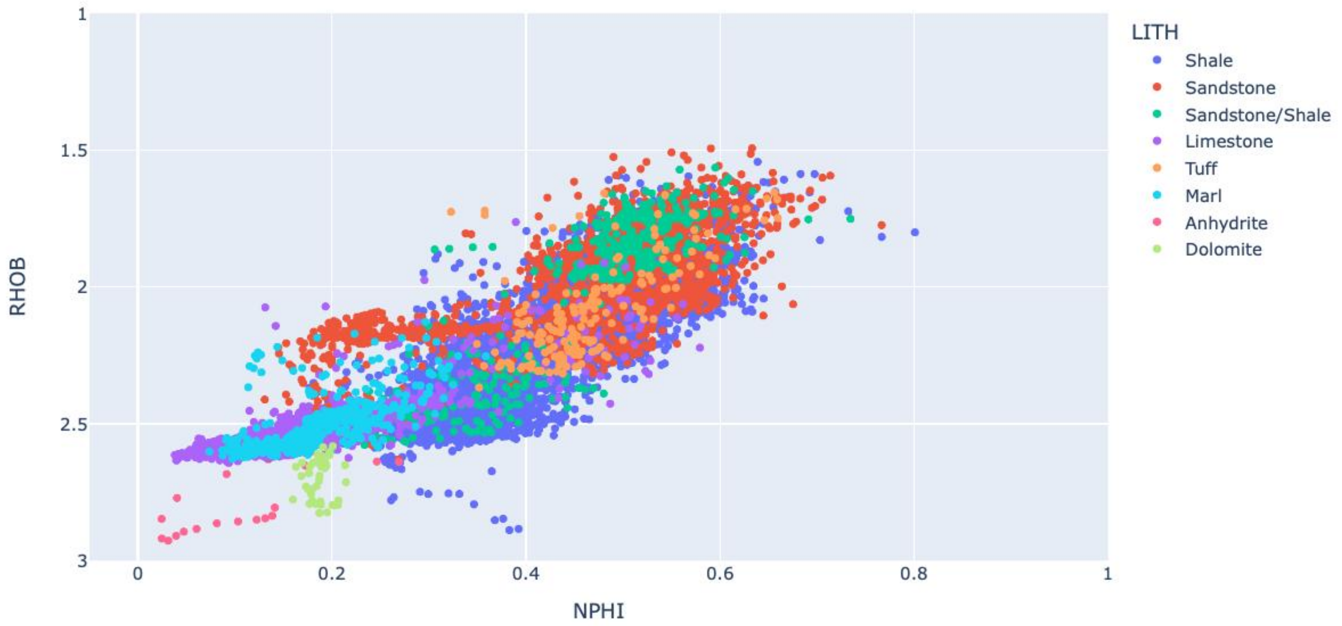
Interactive Scatter Plots

Purpose

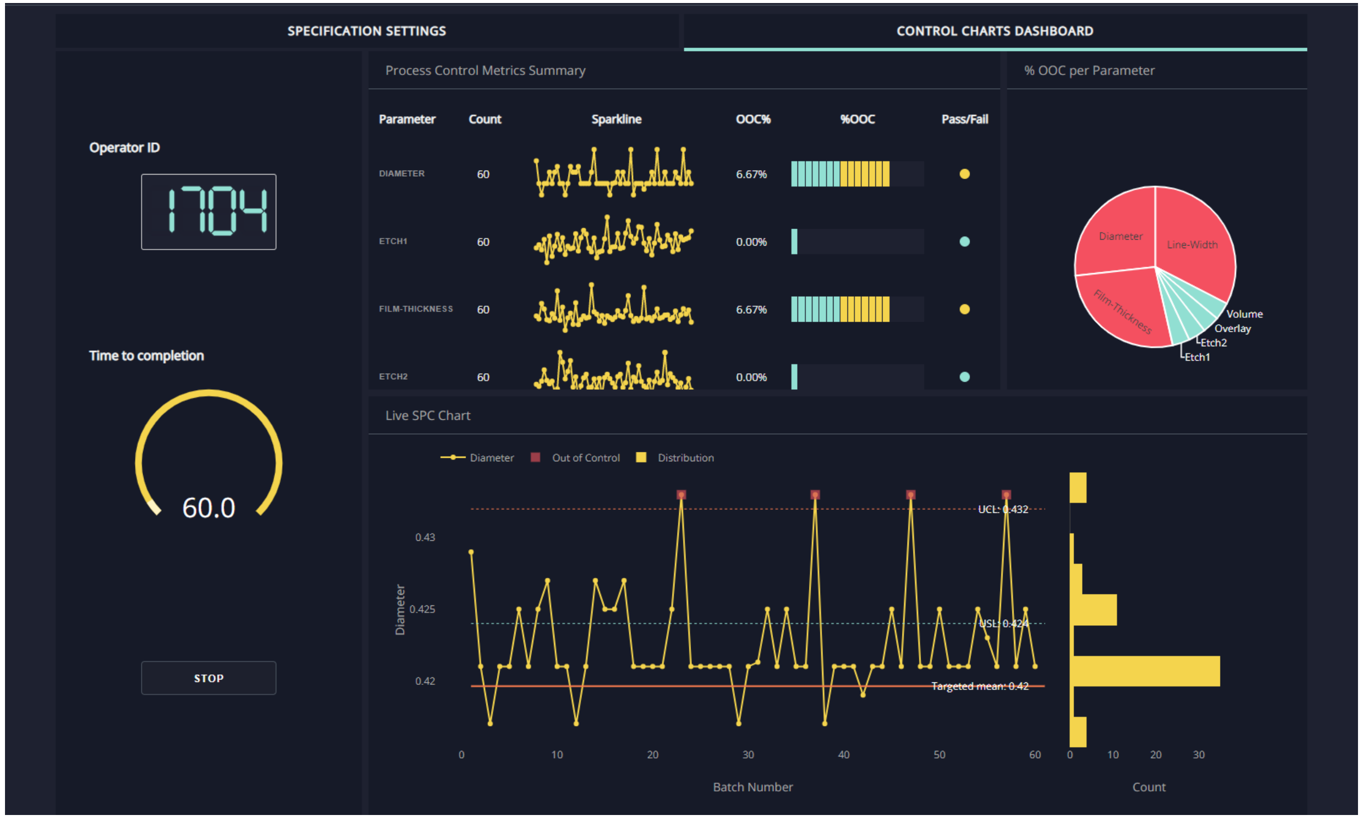
- Explore relationships between variables
- Identify clusters and outliers interactively

Why Interactive Scatter Plots?

- Hover reveals additional dimensions
- Zoom helps analyze dense regions
- Ideal for exploratory analysis



Plotly for Dashboards and Reporting



Applications

- Business intelligence dashboards
- Analytical reports
- Data storytelling presentations
- Interactive dashboards allow stakeholders to **ask their own questions** from the data.

Plotly vs Static Visualization

Aspect	Static Charts	Interactive Charts
User control	No	Yes
Detail on demand	No	Yes
Exploration	Limited	High
Business usability	Moderate	High

Common Mistakes in Interactive Visualization

- Overloading charts with too many interactions
- Ignoring performance for large datasets
- Using interaction where simple charts suffice

Best Practice

- Use interactivity to **enhance understanding**, not to distract.

Static Data Visualization VS Interactive Data Visualization

Factor	Static Data Visualization	VS	Interactive Data Visualization
Trends and patterns	Offer a snapshot view of the data		Enable real-time interaction and exploration
Collaboration	Lacks real-time collaboration capabilities		Facilitate seamless real-time collaboration
Real-time data updates	Data remains static once visualized		Data updates dynamically, reflecting real-time changes
User engagement	Passive viewing experience		Active engagement through exploration and interaction
Decision-making flexibility	Limited ability to adjust views or parameters		Allows for on-the-fly adjustments to analyze different perspectives
Customization options	Fixed presentation style		Customizable views and layouts based on user preferences
Exploring granular details	Provide a fixed view of data		Allows managers to drill down into data points
Manipulating data for scenario analysis	Limited flexibility in analyzing scenarios		Enable real-time manipulation of data parameters
Training and onboarding	Limited learning opportunities with static views		Enhanced learning experience with interactive manipulation and experimentation

5.5 Visualization-Driven Insight Generation

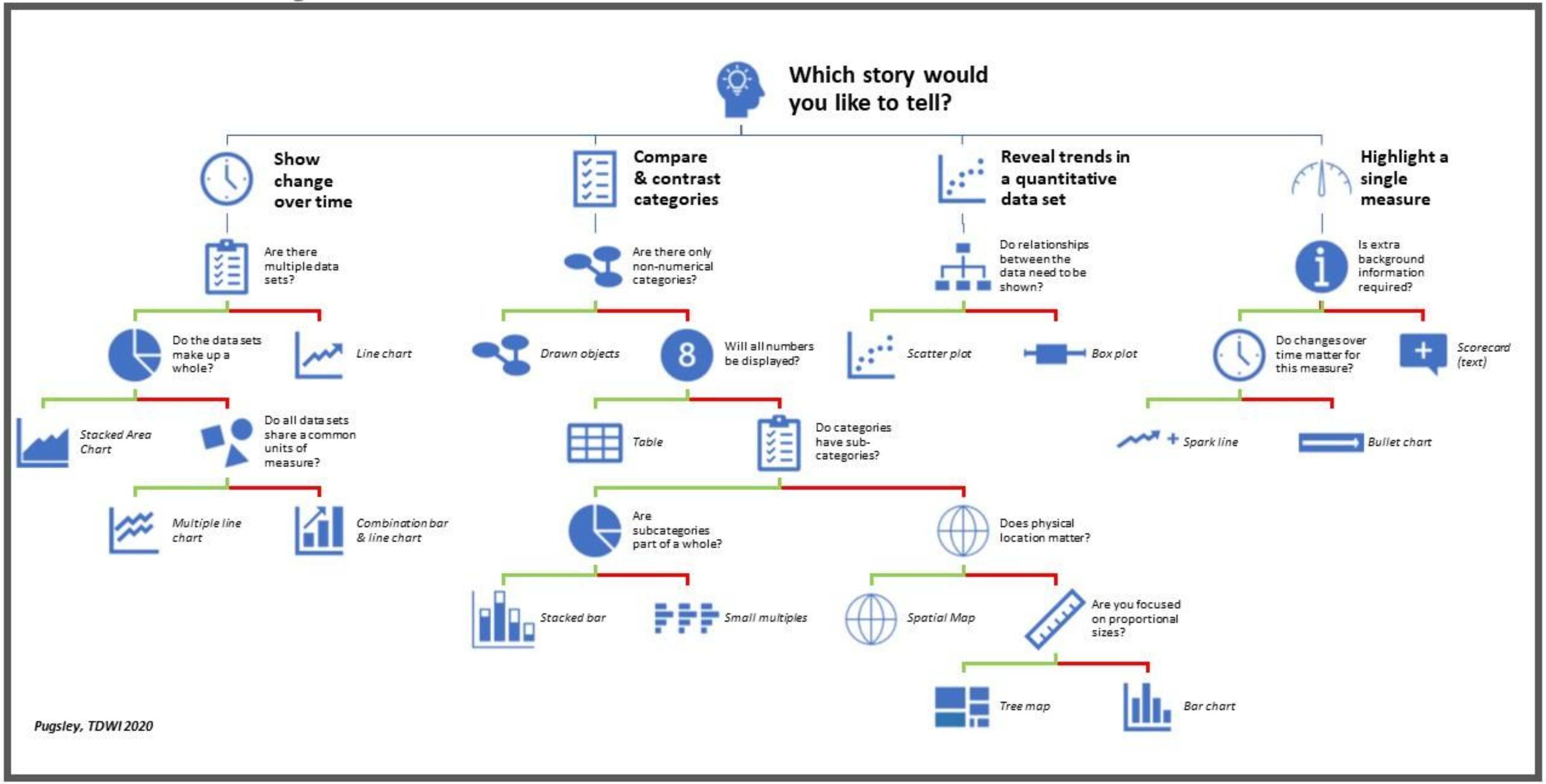
What is Insight Generation?

Insight generation is the process of:

- Interpreting visual patterns
- Understanding *why* those patterns exist
- Converting observations into **actionable conclusions**

Visualization is not the end goal; **insight and decision-making are**.

Data Story Visualization: A Decision Tree



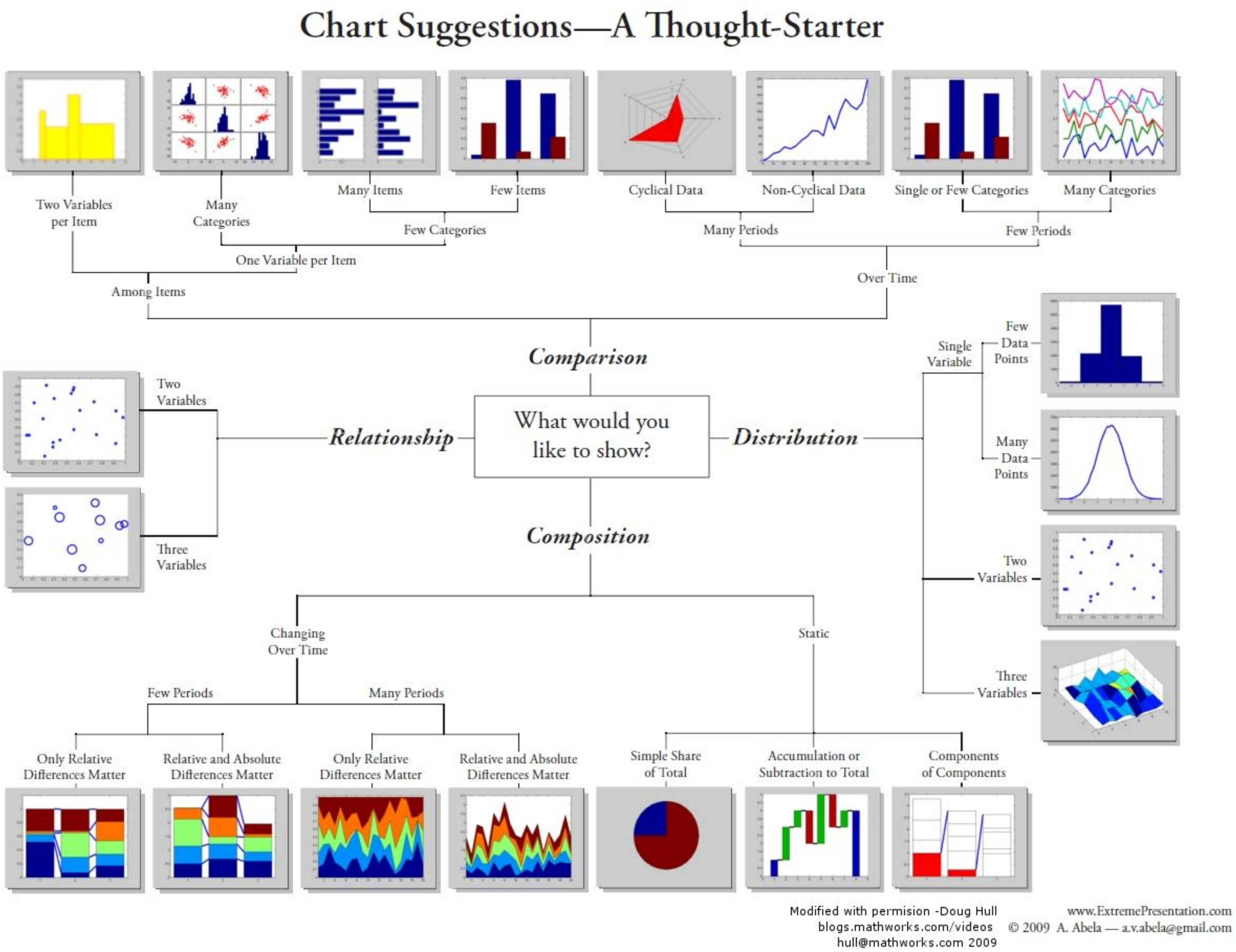


Chart Suggestions—A Thought-Starter
Two Variables per Item
Many Categories
Many Items
Few Items
Few Categories
One Variable per Item
Among Items
Cyclical Data
Non-Cyclical Data
Single or Few Categories
Many Categories
Many Periods
Few Periods
Over Time
Comparison
What would you like to show?
Relationship
Two Variables
Three Variables
Distribution
Single Variable
Few Data Points
Many Data Points
Two Variables
Three Variables
Composition
Changing Over Time
Static
Few Periods
Many Periods
Only Relative Differences Matter
Relative and Absolute Differences Matter
Only Relative Differences Matter
Relative and Absolute Differences Matter
Simple Share of Total
Accumulation or Subtraction to Total
Components of Components
Modified with permission -Doug Hull
blogs.mathworks.com/videos
hull@mathworks.com 2009
www.ExtremePresentation.com
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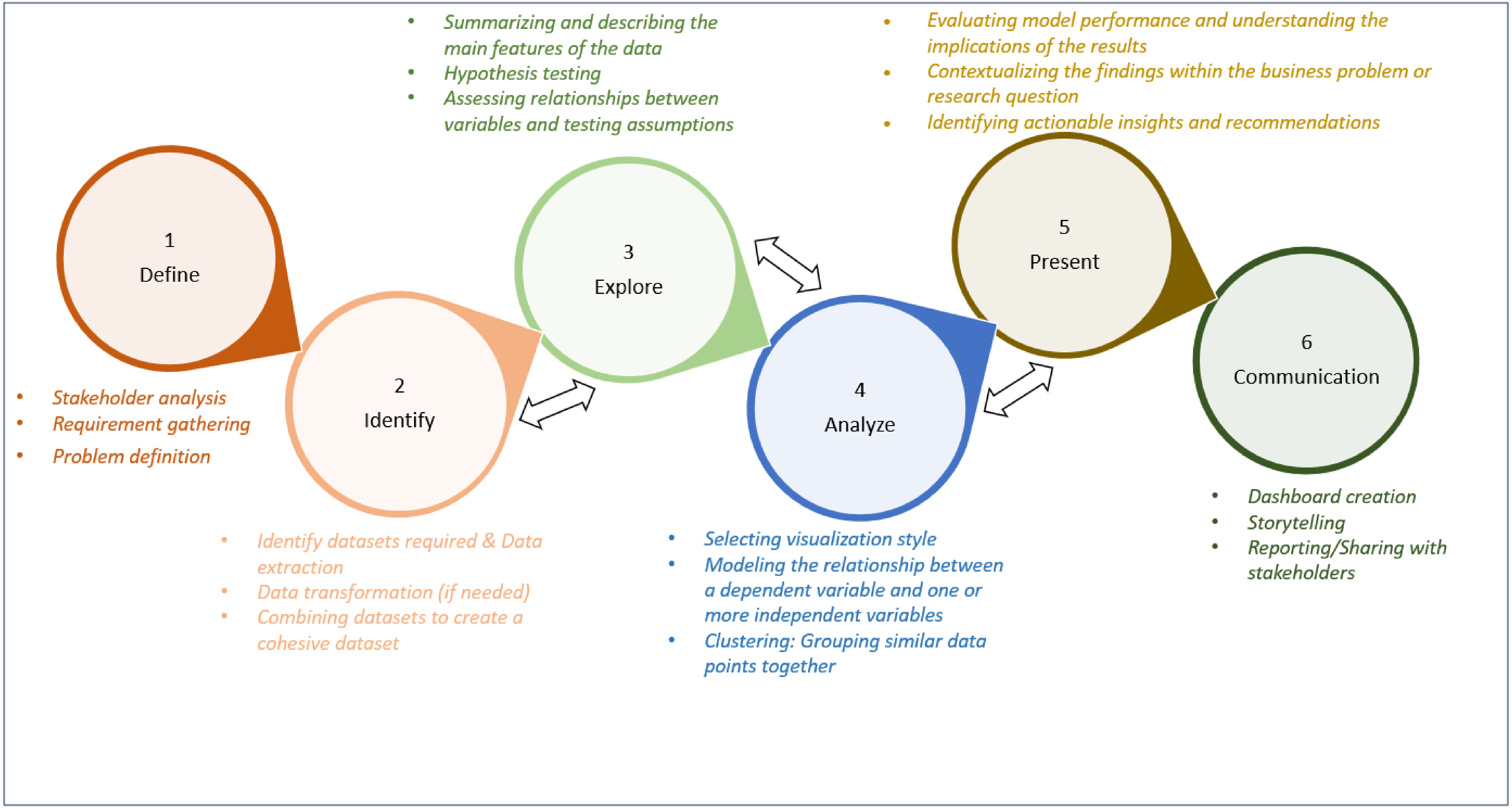
From Data to Insight: The Process

Step-by-Step Flow

1. Raw data is visualized
2. Patterns and trends are observed
3. Questions are raised
4. Insights are derived
5. Decisions or recommendations are made

This process connects **analysis → interpretation → action**.


1
Define
2
Identify
3
Explore
4
Analyze
5
Present
6
Communication
Stakeholder analysis
Requirement gathering
Problem definition
Identify datasets required & Data extraction
Data transformation (if needed)
Combining datasets to create a cohesive dataset
Summarizing and describing the main features of the data
Hypothesis testing
Assessing relationships between variables and testing assumptions
Selecting visualization style
Modeling the relationship between a dependent variable and one or more independent variables
Clustering: Grouping similar data points together
Evaluating model performance and understanding the implications of the results
Contextualizing the findings within the business problem or research question
Identifying actionable insights and recommendations
Dashboard creation
Storytelling
Reporting/Sharing with stakeholders

Identifying Patterns in Visualizations

Common Patterns

- **Trends:** upward or downward movement
- **Outliers:** unusual or extreme values
- **Clusters:** groups of similar observations
- **Gaps:** missing or sparse data regions

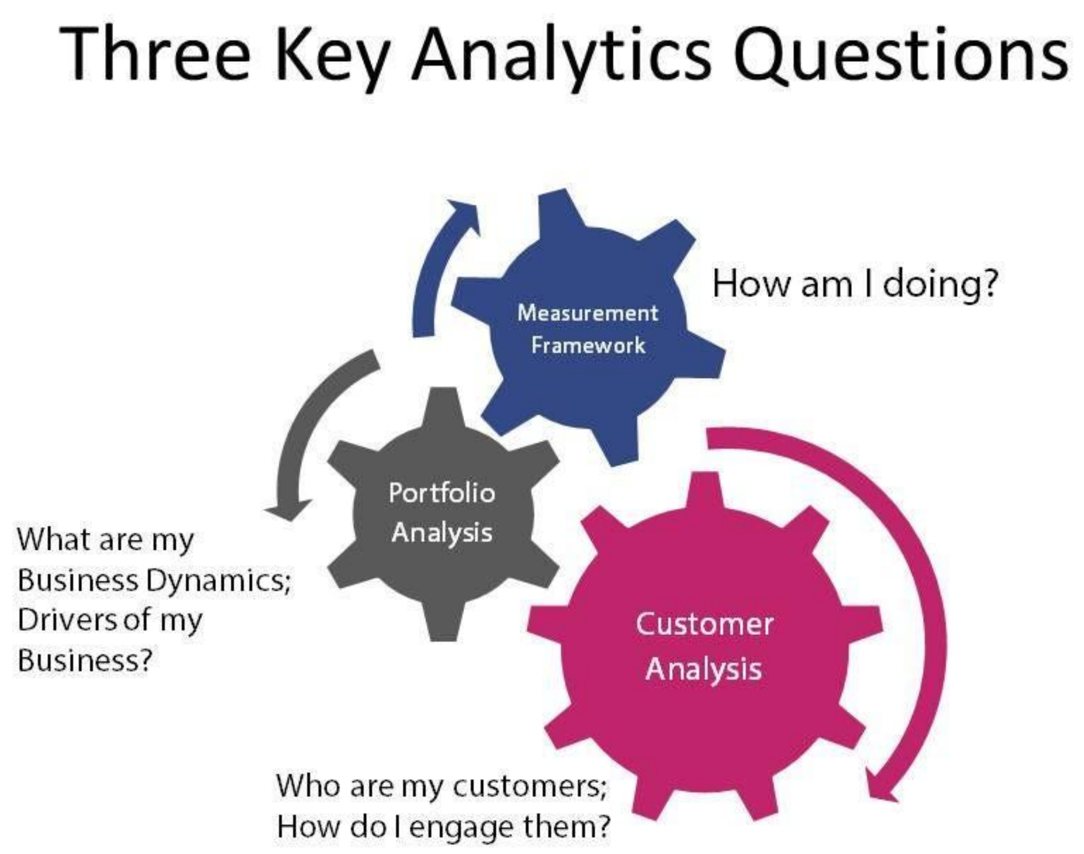
Recognizing patterns is the **first step toward insight**.

Asking the Right Questions

Guiding Questions

- Why is this trend increasing or decreasing?
- What caused this sudden change?
- Are outliers errors or meaningful events?
- Does correlation imply causation?

- Good insights come from **good questions**, not just charts.

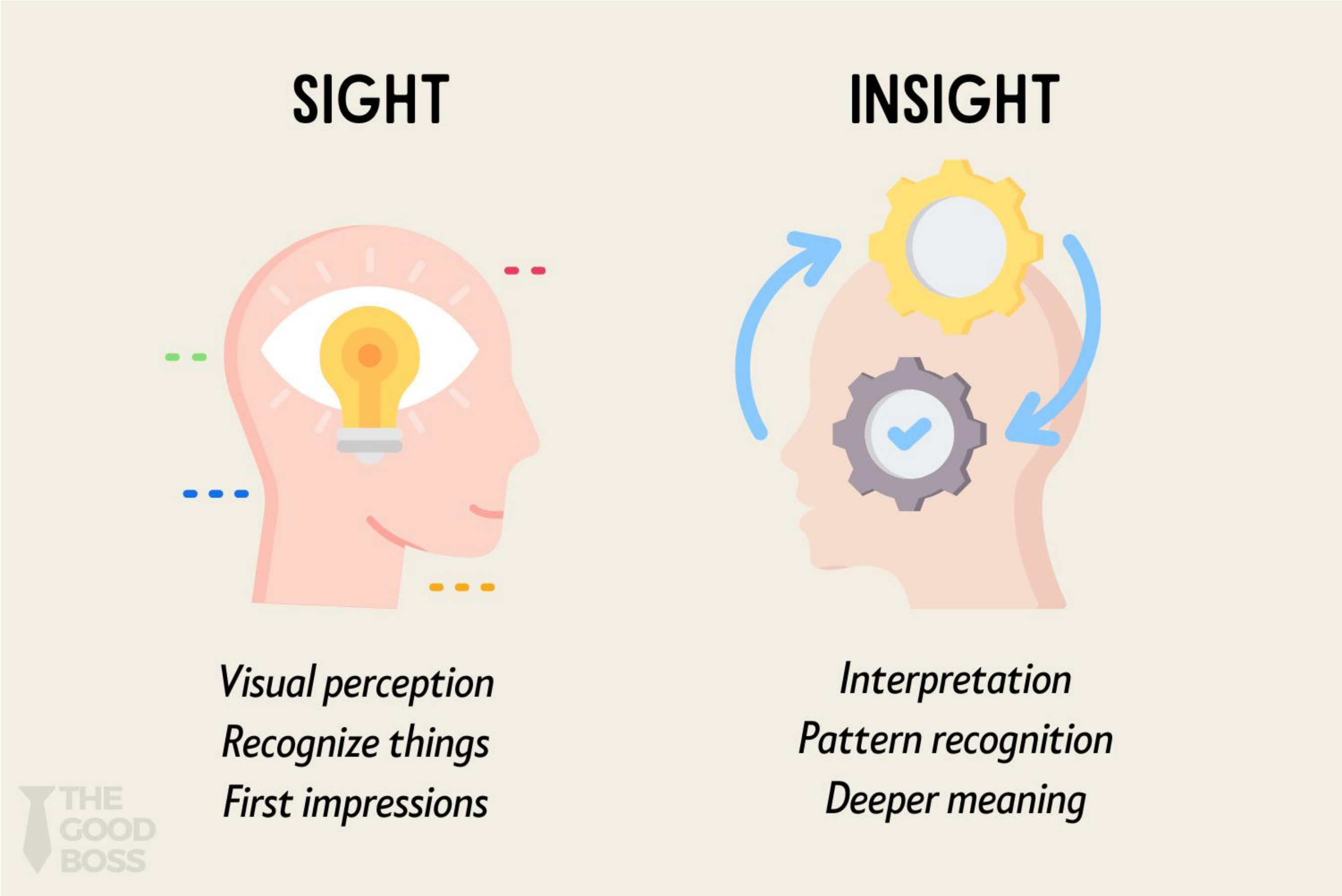


Turning Observations into Insights

Observation vs Insight

- Observation: “Sales dropped in March.”
- Insight: “Sales dropped in March due to supply chain disruption.”

Insights explain **reason and impact**, not just patterns.



Communicating Insights Effectively

Best Practices

- Highlight key findings using annotations
- Keep explanation simple and focused
- Link insights to business or analytical goals

A good insight should be:

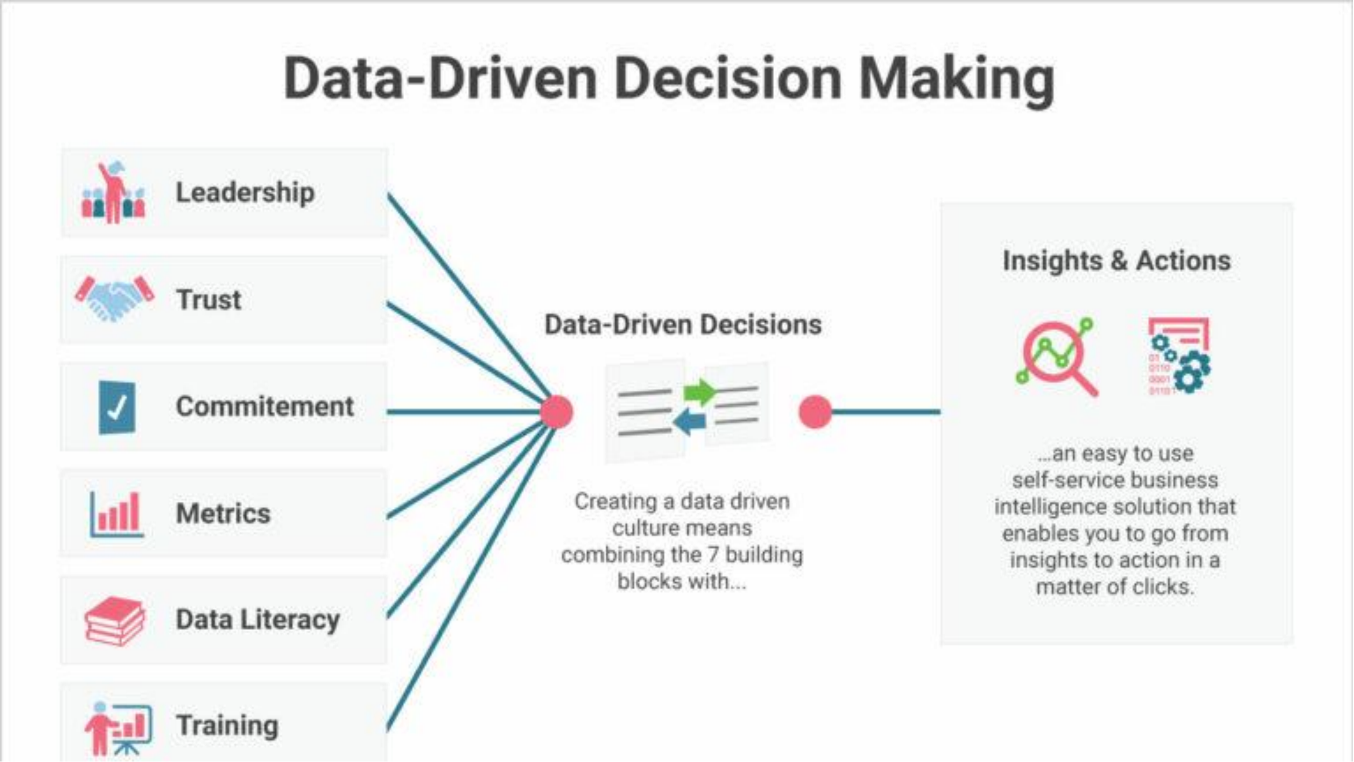
- Clear
- Relevant
- Actionable

Visualization-Driven Decision Making

How Visualization Supports Decisions

- Reduces complexity
- Improves confidence in conclusions
- Enables faster and better decisions

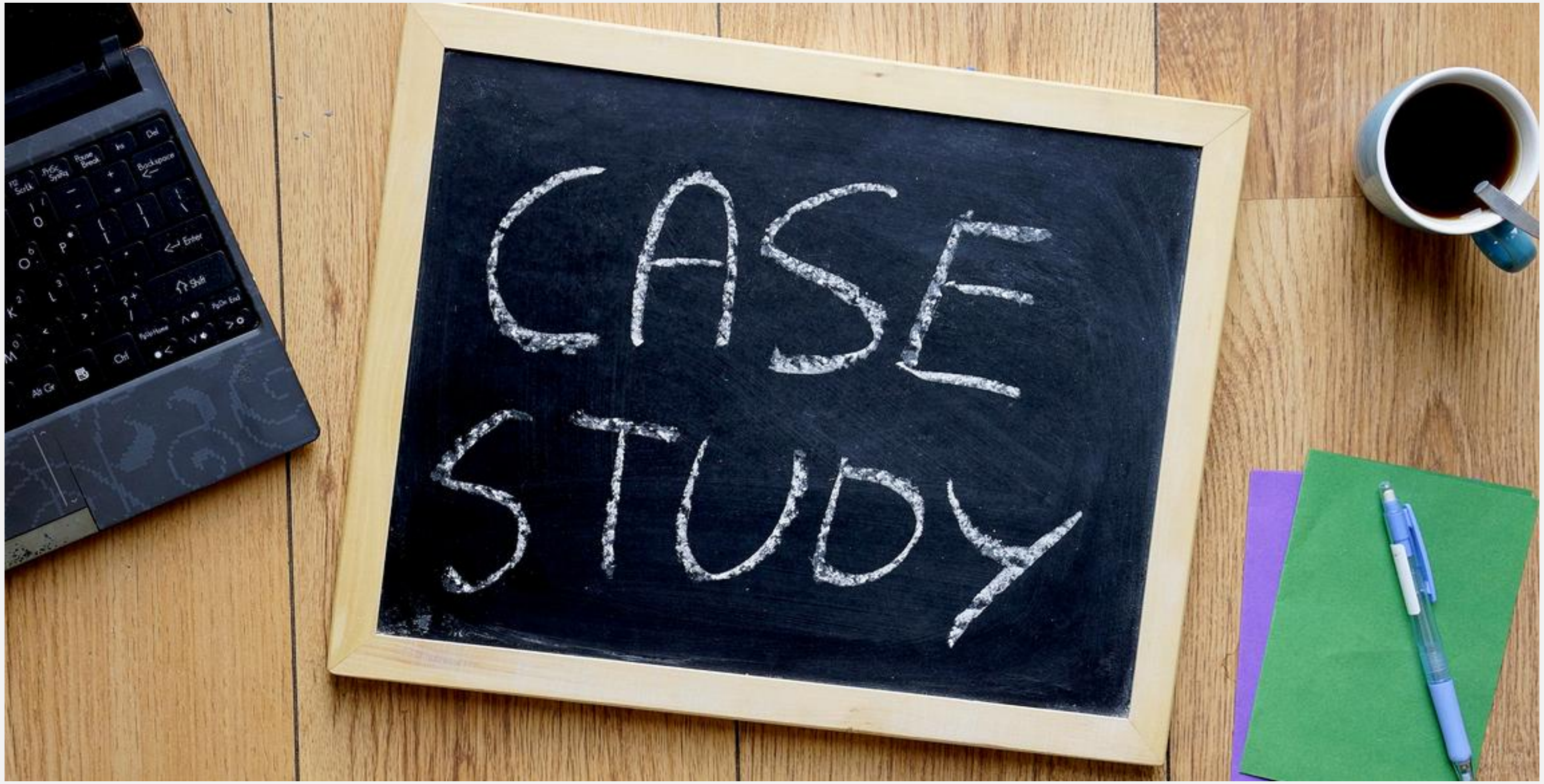
Visualization transforms **data into direction**.



Common Mistakes in Insight Generation

- Describing charts without interpretation
- Ignoring context and domain knowledge
- Assuming correlation implies causation
- Presenting too many insights at once

Focus on **one strong insight per visualization**.



5.6 Case Study: End-to-End Visualization and Reporting Project

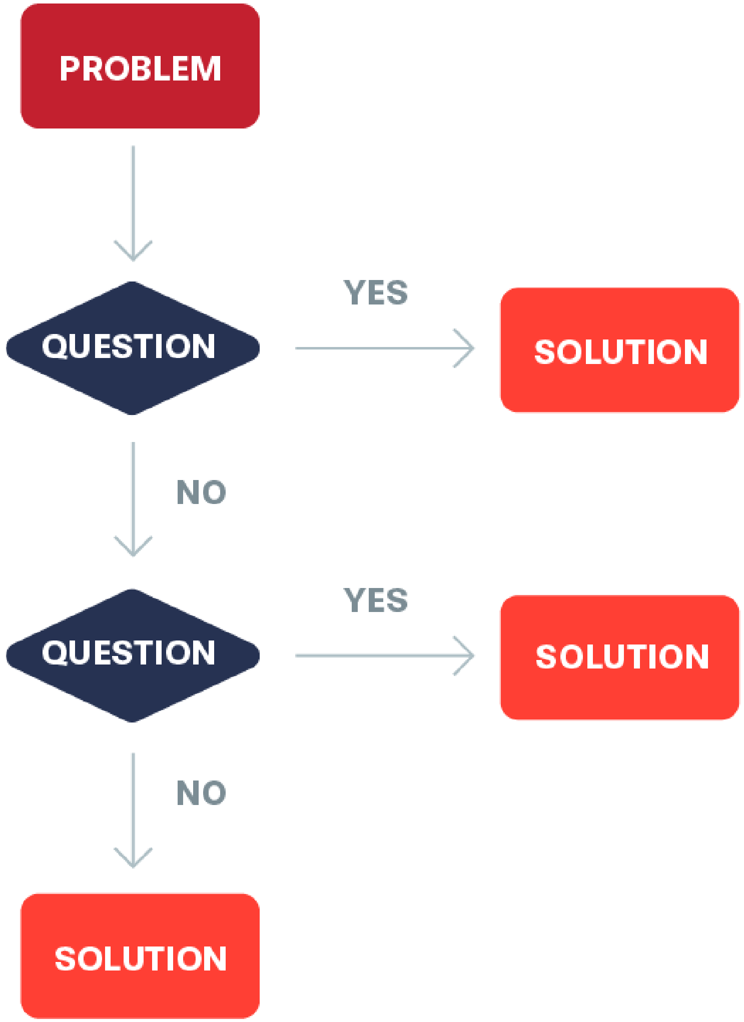
Purpose of the Case Study

The case study demonstrates how to:

- Apply visualization principles in a real scenario
- Select appropriate visualization tools
- Generate insights and communicate results

Objective

- To move from raw data to meaningful insights using visualization.



Overall Workflow of the Case Study

End-to-End Workflow

1. Problem definition
2. Data understanding and preparation
3. Visualization selection and design
4. Insight generation
5. Reporting and presentation

This workflow demonstrates how **raw data is transformed into meaningful insights** using visualization.

Step 1: Problem Definition

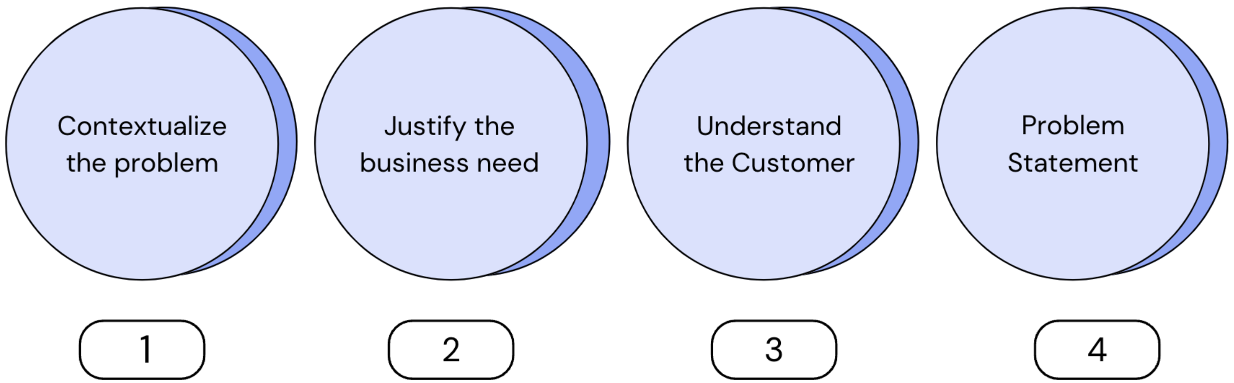
Key Activities

- Understand the business or analytical objective
- Identify key questions to be answered
- Define target audience

Example

- Analyze sales data to identify performance trends

A clear problem definition ensures **focused visualizations**.

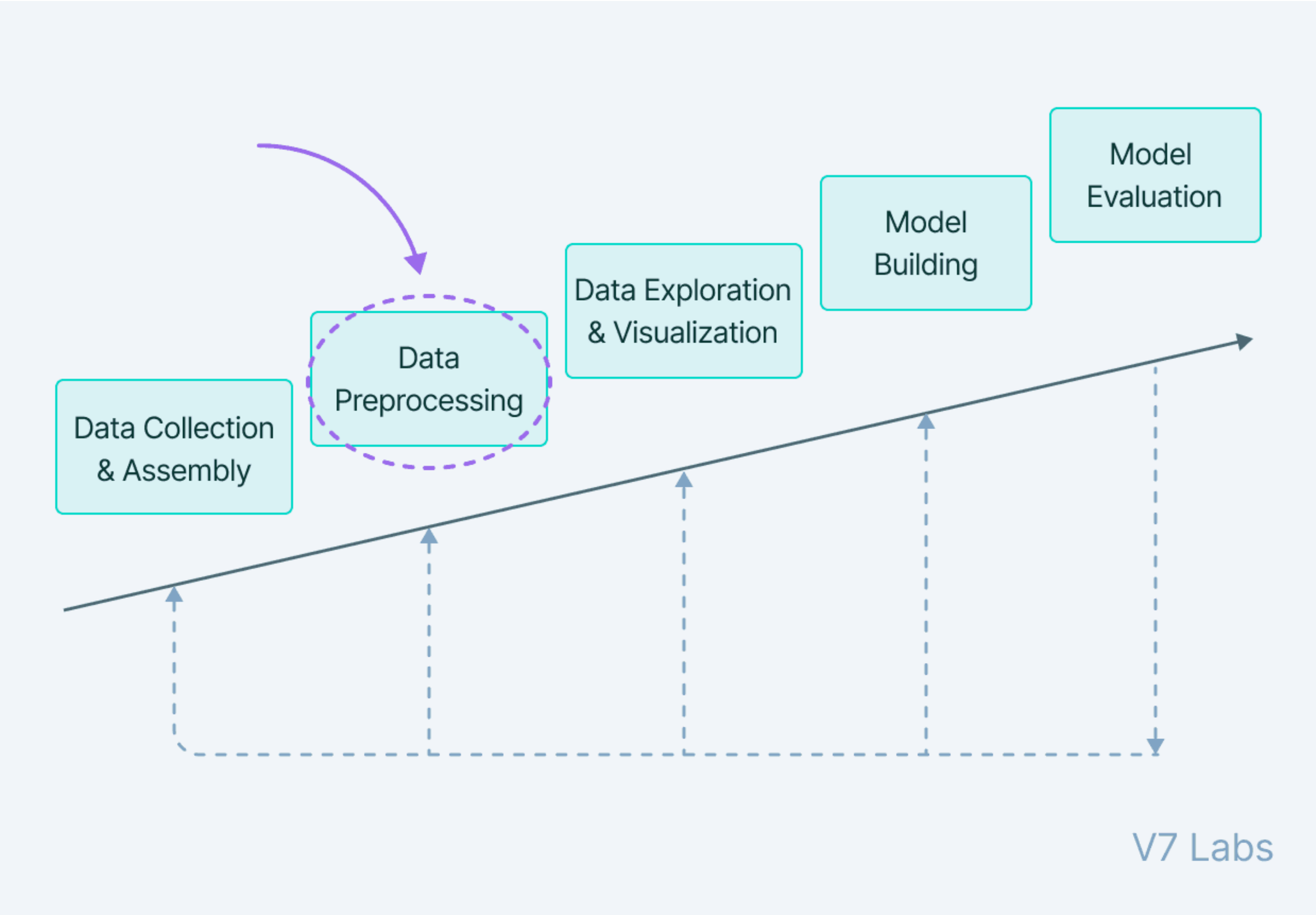


Step 2: Data Understanding and Preparation

Key Activities

- Understand data structure and variables
- Handle missing or incorrect values
- Identify data types (categorical, numerical, time-series)

Clean and well-understood data is essential for **accurate visualization**.



Step 3: Visualization Selection and Design

Decisions Made

- Line plots for trends
- Bar charts for comparisons
- Histograms for distributions
- Scatter plots for relationships
- Heat maps for correlation

Visualization choice must align with **data type and objective**.

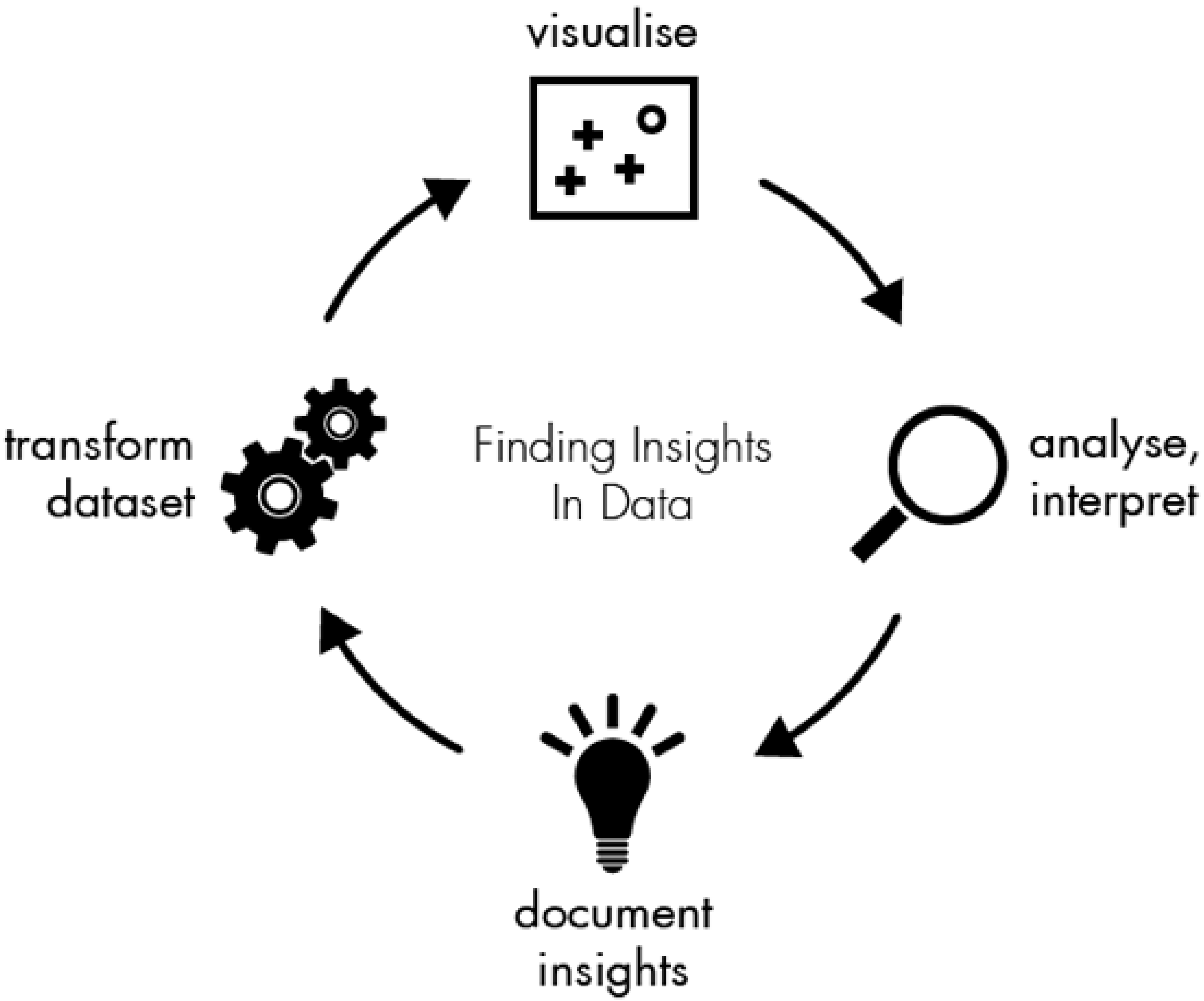
Step 4: Insight Generation

Activities

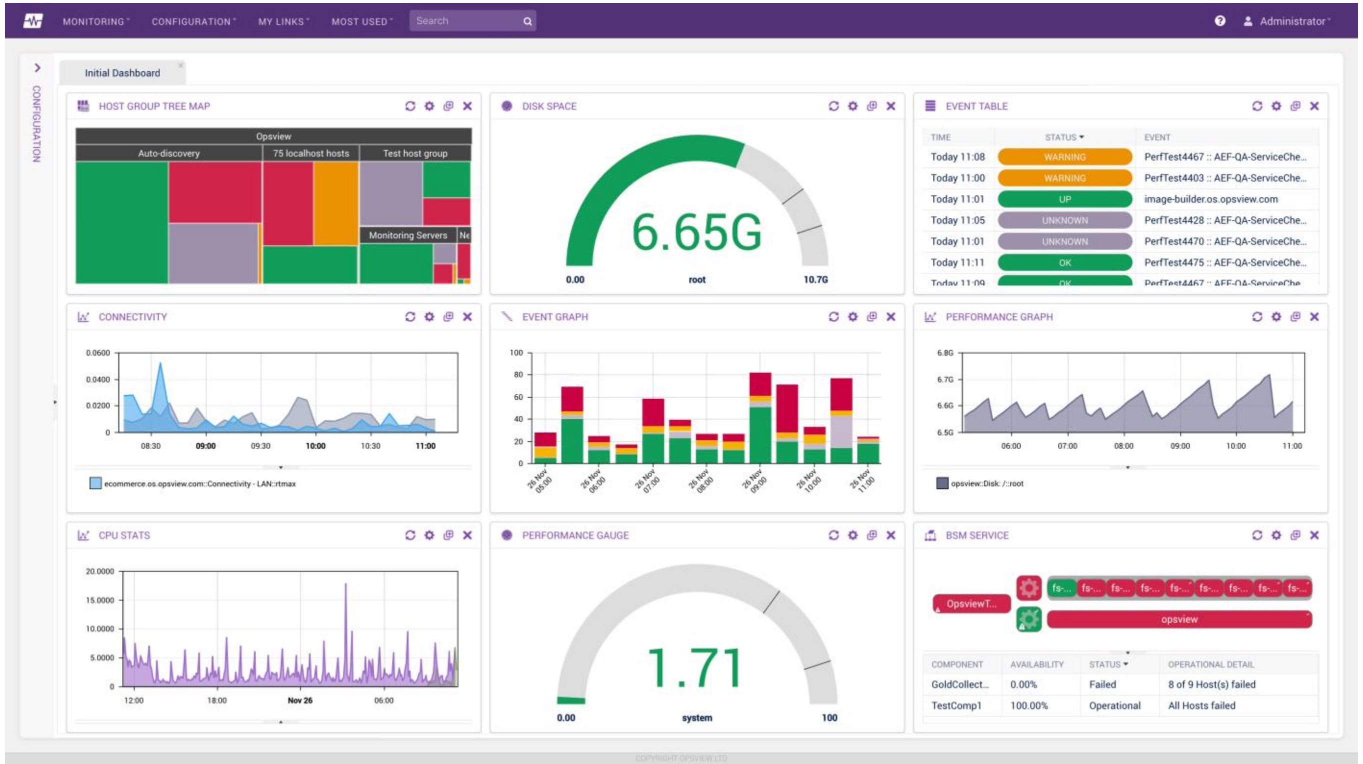
- Identify patterns, trends, and anomalies
- Interpret visual results
- Validate insights with context

Example Insight

- “Sales increased steadily after promotional campaigns.”



Step 5: Reporting and Communication



Deliverables

- Clear charts and dashboards
- Insight summaries
- Actionable recommendations

Reports should be:

- Simple
- Visual
- Decision-oriented

Tools and Visualization Strategy

Tool Usage in the Case Study

- **Matplotlib:** trends, comparisons, basic plots
- **Seaborn:** statistical analysis and EDA
- **Plotly:** interactive charts and dashboards

Each tool is chosen based on **purpose**, not popularity.

3 Data Visualization Tools You Should Know

The Pros & Cons of **MatPlotLib**, **Seaborn**, and **Plotly**

matplotlib

PROS

- Versatile and accessible
- Customizable
- Good documentation
- Universal data viz tool that plugs into many back ends

CONS

- Steep learning curve
- Users need to know Python
- Users need to understand the syntax of Matplotlib, which is based on the software, Matlab

seaborn

PROS

- Quickly creates simple visualizations
- Creates aesthetically pleasing visualizations

CONS

- Limitations on what you can customize
- Visualizations are not as interactive
- Users may need to simultaneously use Matplotlib

plotly

PROS

- Accessible
- Creates aesthetically pleasing visualizations
- Easy to create interactive features within visualization

CONS

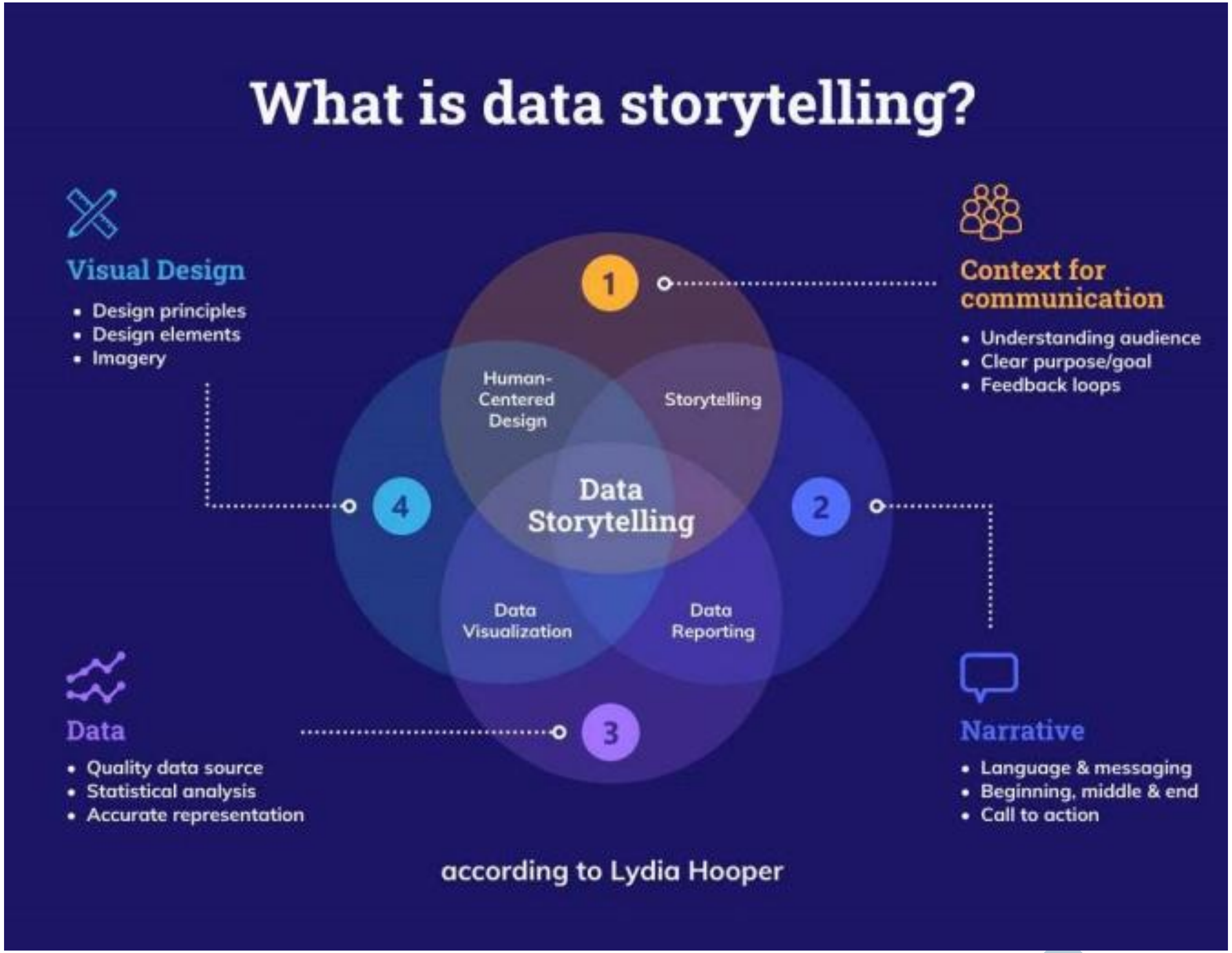
- Steep learning curve
- Users need to know Python
- Uses its own syntax

Learning Outcomes from the Case Study

What This Case Study Demonstrates

- Appy visualization techniques end-to-end
- Integration of multiple visualization libraries
- Conversion of visual patterns into insights
- Effective communication through charts and dashboards

This case study connects **analysis, visualization, and storytelling**.



What is data storytelling?
Visual Design
• Design principles
• Design elements
• Imagery
Context for communication
• Understanding audience
• Clear purpose/goal
• Feedback loops
1
2
3
4
Human-Centered Design
Storytelling
Data Storytelling
Data Visualization
Data Reporting
Data
• Quality data source
• Statistical analysis
• Accurate representation
Narrative
• Language & messaging
• Beginning, middle & end
• Call to action
according to Lydia Hooper

Chapter 5 Recap

VISUALIZATION PRINCIPLES

MATPLOTLIB, SEABORN, PLOTLY

INSIGHT GENERATION

END-TO-END REPORTING

DATA VISUALIZATION TRANSFORMS **DATA INTO DECISIONS**.